

NCSSAR UAS Pilot Classification Overview

Type 3 → Type 2 → Type 1 | OES Designation (Type 1 = Most Qualified)

	Type 3 — Basic	Type 2 — Operational	Type 1 — Advanced
Role	Entry-level pilot. Safe operator in ideal conditions. The student driver.	Reliable search asset. Produces results in moderate complexity. The seasoned driver.	Fully qualified. Deploys in any scenario within waiver limits. Inspires confidence.
Environment	Daylight only, clear weather, calm wind	Degraded lighting, moderate wind, light precip, obstacles	Night, cold/winter, alpine, all conditions within waiver
Terrain	Open, flat to gentle roll	Moderate canopy, rolling foothills, canyon edges	Canyon penetration, heavy canopy, alpine
Navigation	Strict VLOS only	VLOS + approved partial BVLOS	Full BVLOS (1 mi / 200' AGL waiver)
Sensors	Standard camera, basic gimbal	Thermal, IR, zoom, filters, spotlight, speaker	All available, all conditions, rapid switching
VO Coord.	Basic: declare launch, respond to instructions	Active: search strategy, clue reporting, real-time adjustment	Multi-VO network, multi-drone deconfliction
Min. Hours	5 hrs on tracker.kjt.us	10 hrs on tracker.kjt.us	20 hrs on tracker.kjt.us
Defines	Safe aircraft operation + basic SAR integration	Reliable search execution + terrain judgment	Full operational capability within waiver envelope

Progression

Type 3 → Type 2: Add sensor proficiency, moderate terrain experience, VO coordination, and consistent detection performance. Target: a few months after Type 3.

Type 2 → Type 1: Add BVLOS, night ops, telemetry management, multi-VO/multi-drone coordination, and complex terrain mastery. Advance when ready—no fixed timeline.

Shared Requirements (All Types)

- ▶ FAA Part 107 Remote Pilot Certificate (current)
- ▶ Active, trail-ready NCSSAR member in good standing
- ▶ RID2Caltopo qualified (see separate qualification document)
- ▶ Currency: 1 hour of flight per month per drone on tracker.kjt.us
- ▶ IC authorization required for all incident launches
- ▶ All requirements are cumulative—higher types maintain all lower-type standards

Key Principle

Pilot type is an operational qualification. Mission release authority and program governance are separate leadership functions. A Type 1 pilot is the most capable operator—not necessarily the person who decides whether to launch.